Power Budget

Team Number:	105
Project Name:	**N/A**
Team Member Names:	**Joseph, William, Mateo, Sultan**
Version:	**N/A**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors,

All Major Components	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	DAC	MCP4725A0T-E/	2.7–5.5 V	1	0	0	mA
	Op Amp	TLV9001	1.8–5.5 V	1	0.077	0.077	mA
	Electret Microphone	CMC-6027-24T	2–10 V	1	0.5	0.5	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.

+5V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	Op Amp	TLV9001	1.8–5.5 V	1	0.077	0.077	mA
	Electret Microphone	CMC-6027-24T	2–10 V	1	0.5	0.5	mA
	DAC	MCP4725A0T-E/	2.7–5.5 V	1	0	0	mA
						0	mA
						0	mA
	Subtotal					0.577	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					0.72125	mA
c2. Regulator or Source Ch	PIC18F57Q43 Curiosity Nano VBUS	DM164150	5V	1	500	500	mA
	Total Remaining Current Available on +5V Rail					499.27875	mA

C. For each power rail +5V Rail — PIC18F57Q43 Curiosity Nano VBUS (DM164150)

D. Select a specific

External Power Source 1	Component Name	Part Number	yVoltageRan	Outpu	aximumCurre	alCurrent(mA)	Unit
Power Source 1 Selection	USB Power Supply	N/A	5V	5V	500	5000	mA
Power Rails Connected to External Power Source 1	+5V Rail — PIC18F57Q43 Curiosity Nano VBUS (DM164150)		5V				
	Total Remaining Current Available on External Power Source 1					499.27875	mA
